kiseokkeithlee@gmail.com
Google scholar
Website

Linkedin: kiseokkeithlee
X @kiseokmicro / Bluesky

Kiseok Lee

Incoming Assistant professor at Yonsei University, Department of Systems Biology (Mar 2027)
Postdoctoral researcher, Department of Chemical & Biomolecular Engineering,
University of Delaware, Newark, DE, USA
Advisor: Aditya Kunjapur, Co-advisor: Jacques Ravel (University of Maryland, Baltimore)

One-line description:

Microbial ecologist understanding & designing robust microbiome functions.

Education

2025 – 2027	Postdoctoral researcher, University of Delaware, Newark, DE, USA Aditya Kunjapur, Co-advisor: Jacques Ravel (University of Maryland, Baltimore)
2020 – 2025	PhD – Ecology & Evolution, The University of Chicago, Chicago, IL, USA. Advisor: Dr. Seppe Kuehn, Co-advisor: Madhav Mani (Northwestern)
2020 – 2023	Master's – Ecology & Evolution, The University of Chicago / Advisor: Dr. Seppe Kuehn
2018 – 2020	Graduate researcher – Plant Microbiology, Seoul National University (SNU), South Korea. Advisor: Dr. Yong-Hwan Lee
2018	BS – College of Liberal Studies – Biological Sciences (summa cum laude), Economics (summa cum laude), Seoul National University (SNU), Seoul, South Korea.
2010	Posung High School (ranked 1st in ~500 graduating cohort), Seoul, South Korea.

Selected Publications (full list: Google scholar)

Lee, K. K., Liu S., Crocker, K., Wang, J., Huggins, D.R., Tikhonov, M., Mani, M., Kuehn, S. (2025). Functional regimes define soil microbiome response to environmental change. ***Nature***, *644*, 1028-1038 https://doi.org/10.1038/s41586-025-09264-9

Schmitt, M. S., **Lee, K. K**. (co-first author), Landsittel, J. A., Bunbury, F., Vitelli, V., and Kuehn, S. (2026). Learning functional groups in complex microbiomes. bioRxiv (under revision at *Cell Systems*).

Landsittel, J., Howe, A., Mani, M., **Lee, K. K**. (co-corresponding author), and Kuehn, S. Enzyme variants shape metabolic activity in microbial communities (in prep.).

Yousef, M., **Lee, K. K.**, Tang, J., Charisopoulos, V., Willett, R., & Kuehn, S. (2026). Epistatic interactions inform rational design of synthetic microbial communities for bioremediation. ***Nature Microbiology**, 11*, 1995-2007.

Crocker, K., **Lee, K. K.**, Chakraverti-Wuerthwein, M., Li, Z., Tikhonov, M., Mani, M., Gowda, K., Kuehn S. (2024). Environmentally dependent interactions shape patterns in gene content across natural microbiomes. ***Nature Microbiology***, *9*, 2022-2037.

Oliveira, R., Pandey, B., **Lee, K. K.**, Yousef, M., …, Kuehn, S., Raman, A. (2024). Statistical design of a synthetic microbiome that clears a multi-drug resistant gut pathogen. bioRxiv.

Lee, K. K., Park, Y., & Kuehn, S. (2023). Robustness of microbiome function. ***Current Opinion in Systems Biology***, *36*, 100479.

Kim, H., Jeon, J., **Lee, K. K.**, & Lee, Y. H. (2022). Longitudinal transmission of bacterial and fungal communities from seed to seed in rice. ***Communications Biology***, *5*(1), 772.

Lee, K. K., Kim, H., & Lee, Y. H. (2022). Cross-kingdom co-occurrence networks in the plant microbiome: Importance and ecological interpretations. ***Frontiers in Microbiology***, *13*, 953300.

Kim, H., **Lee, K. K**. (co-first author), Jeon, J., Harris, W. A., & Lee, Y. H. (2020). Domestication of Oryza species eco-evolutionarily shapes bacterial and fungal communities in rice seed. ***Microbiome***, *8*, 1-17.

Experience

Postdoc at University of Delaware (PI: Aditya Kunjapur): June 2025 – Present
• Devising synthetic biology solutions to treat recurrent bacterial vaginosis in the vaginal microbiome (co-advisor: Jacques Ravel, University of Maryland, Baltimore)

PhD candidate at University of Chicago (PI: Dr. Seppe Kuehn): Sep 2020 – May 2025
• Investigating the robustness of metabolic activity of soil microbiome in N cycling.
• Excellence: grant writing, project management/leading, collaboration/communication
• Expertise: quantitative thinking, big data analysis (R, Python), ecological modeling, machine learning, sequencing data processing, library prep & operating sequencers, high-throughput microcosm experiments/metabolite measurements, anaerobic cultures.

Research team leader of SymBiome: Jun 2019 – Dec 2019
• Led a research team to optimize the plant microbiome in hydroponics systems to promote romaine growth by securing funding from CJ Blossom Idea Lab (100M won ~$80K/year).

Researcher at SNU (PI: Dr. Yong-Hwan Lee): Sep 2018 – May 2020
• Led a project and published a paper (>150 citations) on the effect of crop domestication on seed microbiomes with seeds of 43 wild and domesticated rice varieties.

Grants & Honors

2026, Nominated for the BSD Best Dissertation Award at the University of Chicago
2026, Awarded E&E Dissertation Award at the University of Chicago
2025, Elected full member, Sigma Xi
2023, Initiated, crafted, and received Community Science Program (CSP FY24) of Joint Genome Institute (JGI) funded by the Department of Energy (DOE) (10 Tb of sequencing, PI: Seppe Kuehn)
2025, Awarded 2025 BSD Conference Awards ($500)
2023, Awarded 2023 Donald Steiner Awards ($2,000)
2022, Awarded KASF (Korean American Scholarship Foundation) scholarship ($2,000)
2022, Awarded Hinds research fund @UChicago Committee of Evolutionary Biology ($2,500)
2022, Awarded KSEA (Korean Scientist Engineer Association)-KUSCO scholarship ($2,000)
2020, Awarded Overseas PhD Scholarship, Korea Foundation for Advanced Studies (KFAS)
2019, Awarded CJ Blossom Idea lab open innovation grant ($252,738/3-year; terminated in 6 months)
2016 – 2017, Undergraduate student scholarship, Korea Foundation for Advanced Studies (KFAS)
2016 – 2017, Dean's List / Full tuition scholarship, Eminence Scholarship in SNU

Talks

2026 Sep 9, Invited talk – Fermentation Club seminar series, Department of Biochemistry and Microbiology, Rutgers, NJ, USA
2026 Apr 16, Invited talk – Department of Chemical Engineering at Ewha Womans University, Korea
2026 Apr 15, Invited talk – Department of Chemical and Biological Engineering at SNU, Korea
2026 Apr 9, Talk – 2026 KSBB Spring Meeting and International Symposium, Yeosu, Korea
2026 Apr 2, Invited talk – Engineering Biology departmental seminar at KAIST, Korea
2025 Aug 29, Invited talk – Biology department seminar at **University of Pennsylvania**, USA
2025 July 5, Talk – **Gordon Research Seminar**: Microbial Population Biology, Andover, NH, USA
2025 May 12, Invited talk – Online seminar at Modeling Ecology Evolution (host: Hye Jin Park)
2025 May 9, Invited talk – Biological sciences department seminar at Sookmyung University, Korea
2025 Apr 30, Invited talk – Biology & Microbiology dept. at Changwon National University, Korea
2025 Apr 29, Invited talk – Plant Microbiology Seminar at Seoul National University, Korea
2025 Apr 29, Invited talk – Microbiology department seminar at Chungbuk University, Korea
2025 Apr 24, Talk – RECOMB-microbiome conference in Seoul, Korea
2025 Apr 23, Invited talk – Biological sciences dept. at Seoul National University, Korea
2025 Apr 18, Invited talk – Biology department seminar at Inha University, Korea

2025 Apr 10, Invited talk – Geology Club Seminar series at **Caltech**, CA, USA
2025 Mar 18, Invited talk – QMicro Seminar at **Ohio State University**, USA
2025 Mar 1, Talk – KSEA research talk at UChicago
2025 Jan 13, Invited talk – Institute for Genome Sciences at University of Maryland, Baltimore, USA
2024 Aug 23, Talk – **ISME**, Cape Town, South Africa
2024, Gave 1-hour talks on soil functional regime 8 times at UChicago: Kronforst lab (Jan 31), Allesina lab (Feb 6), Coleman lab (Feb 8), Pfister/Wootton lab (Feb 9), Raman lab (Feb 15), Pincus lab (Feb 22), CLS chalk talk (Mar 29), Thornton lab (Apr 19)
2023 Apr 1, Talk – SICB (Society for Integrative & Comparative Biology) at UChicago
2023 Mar 30, Talk – TMC23: The Microbiome Center Research Symposium at UChicago
2023 Jan 18, Lightning talk – Space: The Final Frontier of Microbial Communities at Princeton

Posters

2026 Jan 20, Poster – UD Synthetic Biology Symposium 2026 at University of Delaware, USA
2024 Apr 5, Poster – TMC24: The Microbiome Center Research Symposium at UChicago
2023 Dec 9, Poster – ICME23 (International Conference on Microbiome Engineering) at Berkeley

Extracurricular activities

2026 – 2027, Co-president of Korean Graduate Student Association at University of Delaware
2021 – 2024, Co-Chair of Travel Award evaluation committee in *BSD Deans Council* at The University of Chicago (UChicago)
2023 – 2024, Planned & organized *TMC23* (2023 Mar 30) and *TMC24* (2024 Apr 5): The Microbiome Center Research Symposium with UChicago, MBL, Argonne National Laboratory
2023 Jul 26 – Aug 8, Cold Spring Harbor Synthetic Biology Course 2023 (Instructors: Vincent Noireaux, Eric Young, Chase Beisel, Christian Cuba Samaniego, Karmella Haynes)
2023 Jan 12, Planned & organized the first *KSEA Science Conference* at UChicago
2021 – 2022, President of the Korean Graduate Student Association (KGSA) at UChicago

Teaching/Mentorship

2025, 4-hr CLS training to become a mentor: “Effective communication and aligning expectations”
2025, 6-hr MyChoice running a research group training: “Grants, lab management, and hiring/firing”
2024, TA for Theoretical ecology (Instructor: Greg Dwyer, Stefano Allesina)
2022, TA for Quantitative microbial ecology (Instructor: Seppe Kuehn)
2022 – 2023, Field trip TA for Ecology & conservation (Instructor: Cathy Pfister, Eric Larsen)
2026, Mentored Iris Zhang (undergraduate Summer Scholars) for vaginal microbiome engineering
2025 – 2026, Mentored Defne Elbeyli (undergraduate researcher) for ecology in hydrogel project
2025, Mentored Laura Kirshenbaum (rotating PhD student) for soil nutrient niche project
"*Kiseok quickly helped me adapt to the lab and feel welcomed during my rotation. He set me up well for my project while ensuring I was involved in its design and analysis, and I learned a great deal from his guidance.*"
2025, Mentored Cynthia Wang (rotating PhD student) for metagnomics/DNRA project
2023 – 2025, Mentored Jojo Wang (undergraduate researcher) for the pH niche of denitrifiers project
"*Throughout my undergraduate research experience Kiseok was a consistent source of support and clear communication, equipping me to advocate for my own research interests.*" - Jojo
2022 – 2023, Co-mentored Jonathan Tang (undergraduate researcher) for the BPA degrader project
"*Kiseok is a kind and supportive mentor who fostered my interest in research by encouraging me to take the lead in asking questions I was interested in. I could always rely on him to make time for any questions.*" - Jonathan

Academic services: referee for 10+ papers

2026, Invited as a referee at *Environmental Science & Technology*, etc.
2025, Invited as a referee at *Proceedings B*, *Mycobiology*, *Biodegradation*, *BMC genomics*, etc.
2025, Invited as a referee at *Microbiome* journal by the editor (Lynn Schriml)
2024, Invited as a referee at *Nature Communications* by the editor (Cesar Sanchez)
2024, Invited as a referee at *eLife* by the editor (Anne-Florence Bitbol)

2023, Invited as a referee at *Nature Ecology & Evolution* by the editor (Simon Harold)

News outlet

Functional Regimes Shape Soil Microbiome Response–ScienMag, 7/17/2025
Simple rules govern soil microbiome responses to environmental change–The Microbiologist, 7/16/2025
Soil Microbiome Shifts Guided by Basic Environmental Rules–Mirage News, 7/17/2025
Simple Models Reveal Soil's Complex Microbial Conversations–Azo Life Sciences, 7/18/2025
Simple Rules Govern Soil Microbiome Responses to Environmental Change–UChicago News Biological Sciences Division, 7/16/2025
Simple Rules Govern Soil Microbiome Responses to Environmental Change–Northwestern McCormick School of Engineering News, 7/20/2025
単純なルールは、環境の変化に対する土壌微生物叢の反応を支配します–Nipponese news, 7/25/2025
Simple Rules Govern Soil Microbiome Responses to Environmental Change–Newswise, 7/25/2025
A Simple Model Describes How the Soil Microbiome Responds to Environmental Change–Technology Networks, 7/31/2025

References

Dr. Aditya Kunjapur (Postdoc PI)	University of Delaware, Associate Professor of Chemical & Biomolecular Engineering	kunjapur@udel.edu
Dr. Jacques Ravel (Postdoc co-PI)	University of Maryland, Baltimore, John L. Whitehurst Professor of Medicine, Microbiology, and Immunology	jravel@som.umaryland.edu
Dr. Seppe Kuehn (PhD PI)	The University of Chicago, Associate Professor of Ecology and Evolution	seppe.kuehn@gmail.com

"*Kiseok (as a grad student) operates at the level of a postdoc.*" – words from Seppe

Dr. Madhav Mani (PhD co-PI)	Northwestern University, Associate Professor of Engineering Sciences and Applied Mathematics	madhav.mani@gmail.com

"*The project we did together is by far the best one I've done in my career*" – words from Madhav

Dr. Mikhail Tikhonov (Project co-PI)	Washington University in St. Louis, Associate Professor of Physics	tikhonov@wustl.edu
Dr. Cathy Pfister (PhD committee chair)	The University of Chicago, Professor of Ecology and Evolution	cpfister@uchicago.edu
Dr. Karna Gowda (Lab alumni)	The Ohio State University, Assistant Professor of Microbiology	karna.gowda@gmail.com
Dr. Yong-Hwan Lee	Seoul National University, Professor of Agricultural Biotechnology	yonglee@snu.ac.kr